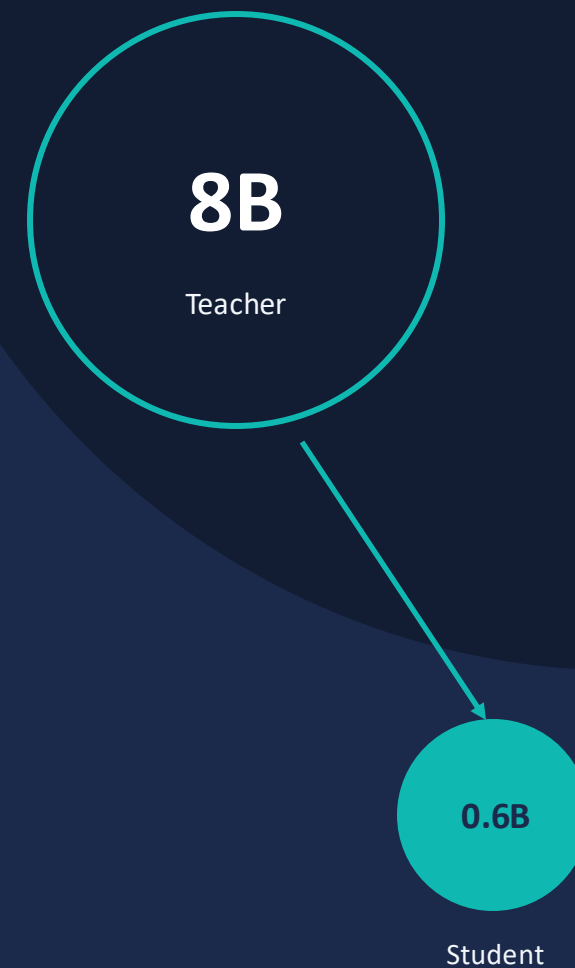


Model Miniaturization

Compressing an 8B Medical-Triage LLM Into a
Deployable 0.6B Assistant

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Applied Artificial Intelligence Lab · Summer Semester 2026



13× smaller · beats the teacher

What We'll Cover



Motivation & Background

Why triage needs AI help, and why that model has to be small

SPEAKER 1



Project Overview

The end-to-end pipeline and two parallel approaches

SPEAKER 1



Methods

Pruning, knowledge distillation, and data-centric LoRA fine-tuning

SPEAKER 1



Experiments & Results

How we evaluated it, what worked, and the trade-offs

SPEAKER 2



Limitations, Future Work & Takeaways

What didn't work, what's next, and the big lessons

SPEAKER 2

Q&A immediately follows — 5 minutes, questions held until the end.

Why This Project? Why a Smaller Model?



The Clinical Problem

- **Triage decides who gets seen first** — in a busy ER, a wrong call can cost minutes that matter.
- **Large medical LLMs reason well** (the project's teacher model is an 8B-parameter biomedical LLM) but are too heavy to run in routine or offline clinical settings.
- **The goal:** keep the clinical reasoning quality, shed almost all of the size.



The target: 8B parameters → well under 1 GB VRAM, with reasoning quality preserved.



Why Miniaturization Matters



Latency

Real-time triage can't wait on a slow round-trip to a data-center GPU.



Memory & cost

Not every clinic can host a multi-GPU inference server.



Edge & offline use

Rural or low-connectivity settings need models that run locally.



Privacy

On-device inference keeps patient data from ever leaving the room.

Background: Four Ways to Shrink a Model



Pruning

Remove less-important weights, attention heads, or whole layers from an already-trained network.



Quantization

Store weights in fewer bits — e.g. 4-bit instead of 16/32-bit floating point — to shrink memory footprint.



Knowledge Distillation

Train a small “student” model to mimic a large “teacher” model's output probabilities, not just hard labels.



LoRA / PEFT

Freeze the base model entirely; train only small, low-rank “adapter” matrices bolted onto it.



This project actually uses all four — a working comparative study, not just theory.

The End-to-End Pipeline

OpenBioLLM-8B
teacher model

Synthetic Training Data
5,100 reasoning samples

used to train the Qwen3-0.6B student in both approaches below

APPROACH 1 · COMPRESSION-CENTRIC

OpenBioLLM-8B

Structured Pruning
heads + layers

Distillation
learns from teacher

Compressed Student
pruned + distilled

APPROACH 2 · DATA-CENTRIC

Qwen3-0.6B
student base

Merge Datasets
synthetic + real (MIMIC, fedmml)

LoRA Fine-Tune
r=16, 4-bit NF4

Fine-Tuned Student
best performer

both outputs → scored on the same 4-dataset, 3-metric evaluation harness

Two Independent Approaches, Compared Head-to-Head



Approach 1

Compression-Centric

- Structured pruning of attention heads and transformer layers
- Recovery Fine Tuning
- Knowledge distillation from the 8B teacher's soft labels
- Goal: compress the architecture itself



Approach 2

Data-Centric

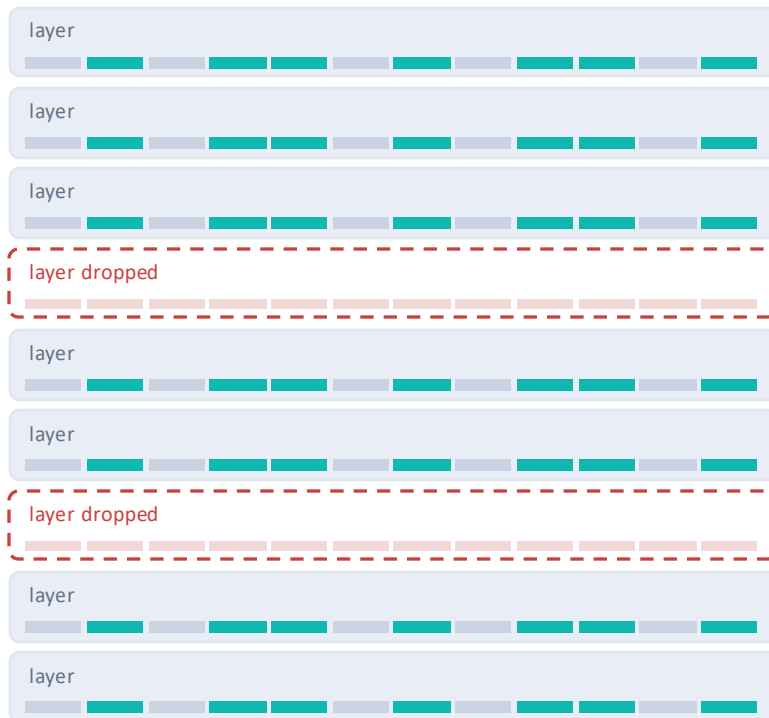
- Synthetic + real clinical data, merged and quality-checked
- LoRA fine-tuning applied to the untouched student
- Goal: better data and training, same architecture



The project's real contribution is this comparison — not any single model.

Method 1: Structured Pruning

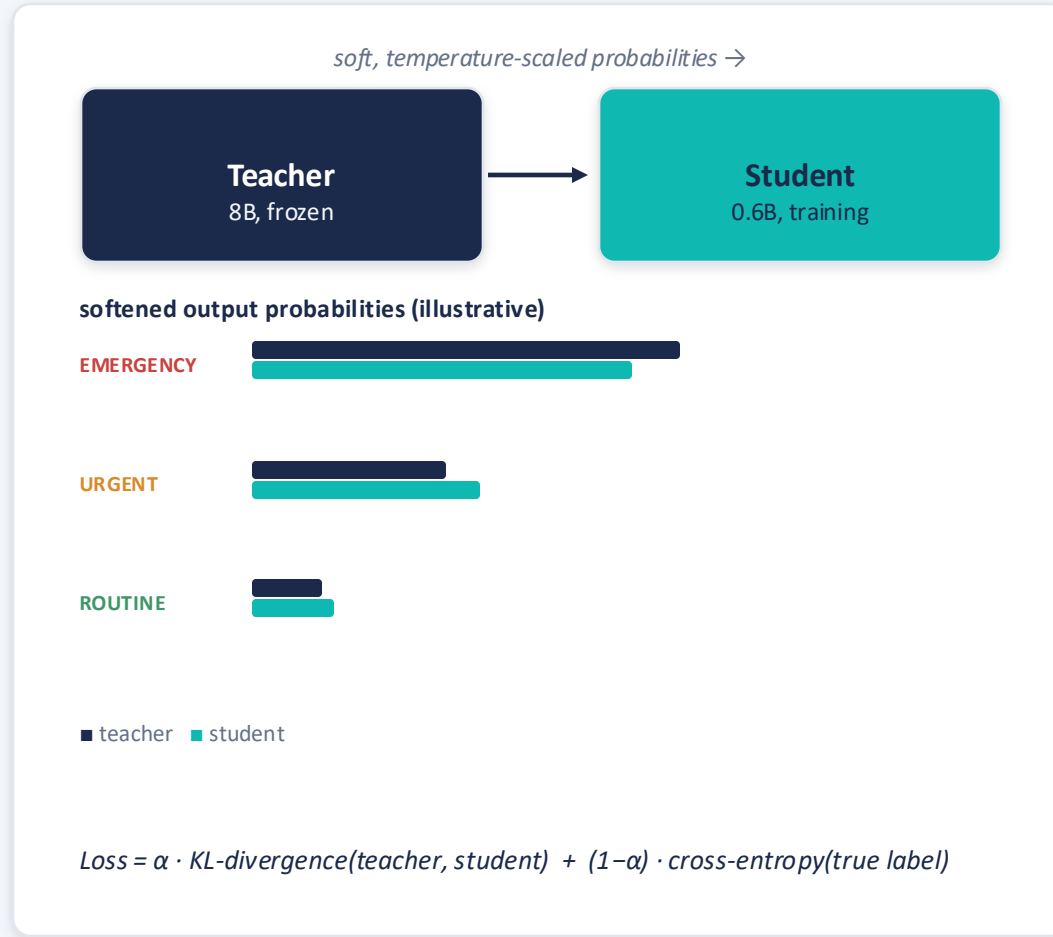
28 transformer layers, 16 heads each



solid tick = kept head · light tick = pruned head (bottom 40%)

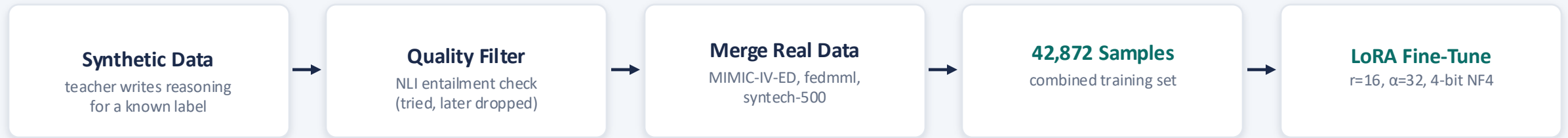
- **Score importance** of every attention head and layer from activation statistics on a small batch of data.
- **Prune heads:** remove the lowest-scoring 40% of attention heads, per layer.
- **Drop layers:** remove 5 whole middle layers (out of 28) by fixed index.
- **Recover:** a short LoRA fine-tune afterward to regain lost accuracy.
- **Code:** src/pruning/ — three focused scripts (~150 lines total).

Method 2: Knowledge Distillation



- **Soft labels carry more signal:** the teacher's full probability spread teaches the student “how confident,” not just “which class.”
- **Combined loss:** weighted mix of KL-divergence to the teacher and cross-entropy to the true label.
- **Two variants trained:** distilling into the plain student, and separately into the pruned student base.
- **Two trainer versions** exist in the repo (v1 → v2) — more on why in the results.

Method 3: Data-Centric LoRA Fine-Tuning



- **Oracle-label prompting:** the teacher is told the correct triage level and writes the clinical justification for it — clean, well-labeled synthetic data at scale (5,100 samples).
- **Real clinical data mixed in:** MIMIC-IV-ED (207 ER visits), fedmml-ed-triage (87K multi-country ER visits), syntech-500.
- **LoRA touches 1.7% (10M trainable / 0.62B = 1.67%) of parameters** — cheap to train, easy to swap adapters without re-training the base model.
- **4-bit quantized base during training** (QLoRA-style) keeps the whole fine-tune runnable on a single GPU.

Datasets Inventory

Dataset	Type	Size	Real/Synthetic	Role
Your synthetic	Generated	5,100	Synthetic (teacher)	Primary training
syntech-ai/medical-triage-500	Downloaded	500	Synthetic-origin	Training mix
MIMIC-IV-ED demo	Real ED	207	Real patients	Training + v3 probe
fedmml (Denmark+Turkey)	Real ED	38,085	Real patients	Training augmentation
fedmml (Latvia)	Real ED	3,000	Real patients	Held-out test
qiaojin/PubMedQA	Biomedical QA	300	Real literature	Reasoning quality check only

How We Evaluate: Datasets & Metrics



Four Evaluation Tiers

- **In-distribution test**

510 held-out synthetic samples

SANITY CHECK

- **Domain-shift test**

3,000 real patients from Latvia, unseen at train time

REAL DATA

- **External real-world test**

42 held-out MIMIC-IV-ED patients

THE HONEST ONE

- **Reasoning-quality check**

BERTScore vs. PubMedQA text

QUALITY ONLY



Accuracy

Overall share of correct triage calls across all three classes.



Macro F1

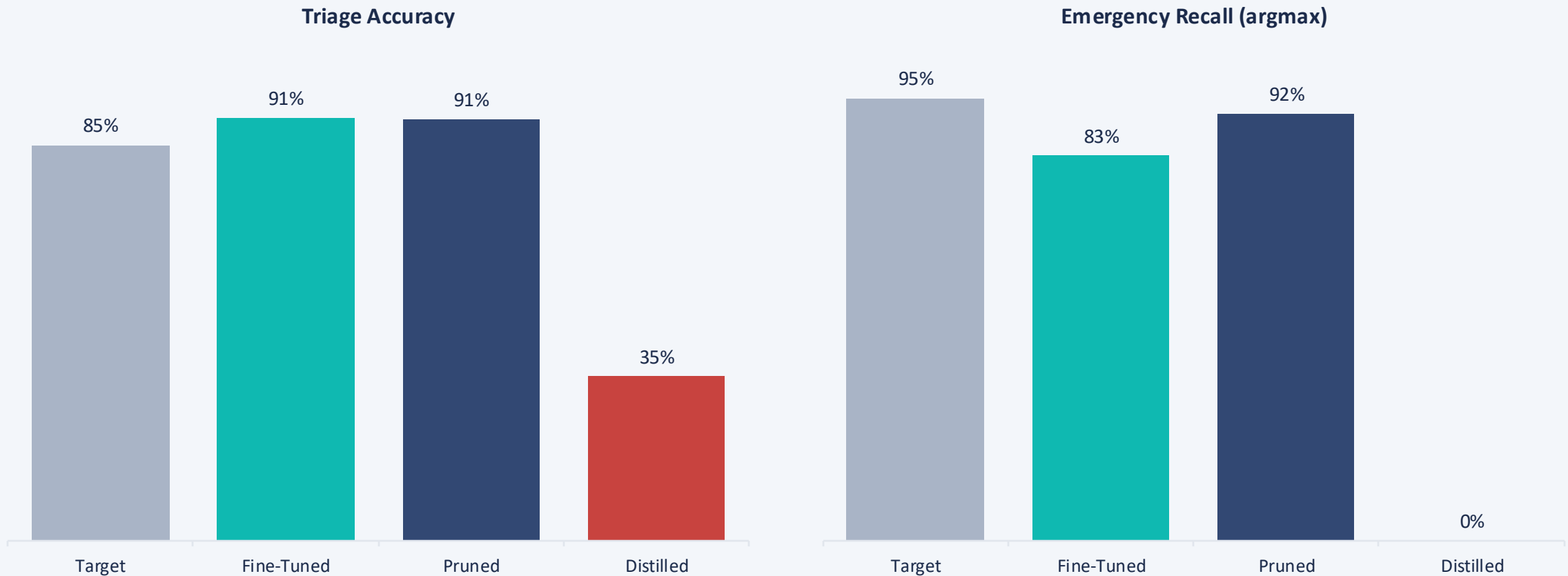
Balances **performance** evenly across EMERGENCY, URGENT, and ROUTINE — rare classes count fully.



Emergency Recall

Of all true emergencies, how many did we actually catch? The safety-critical number.

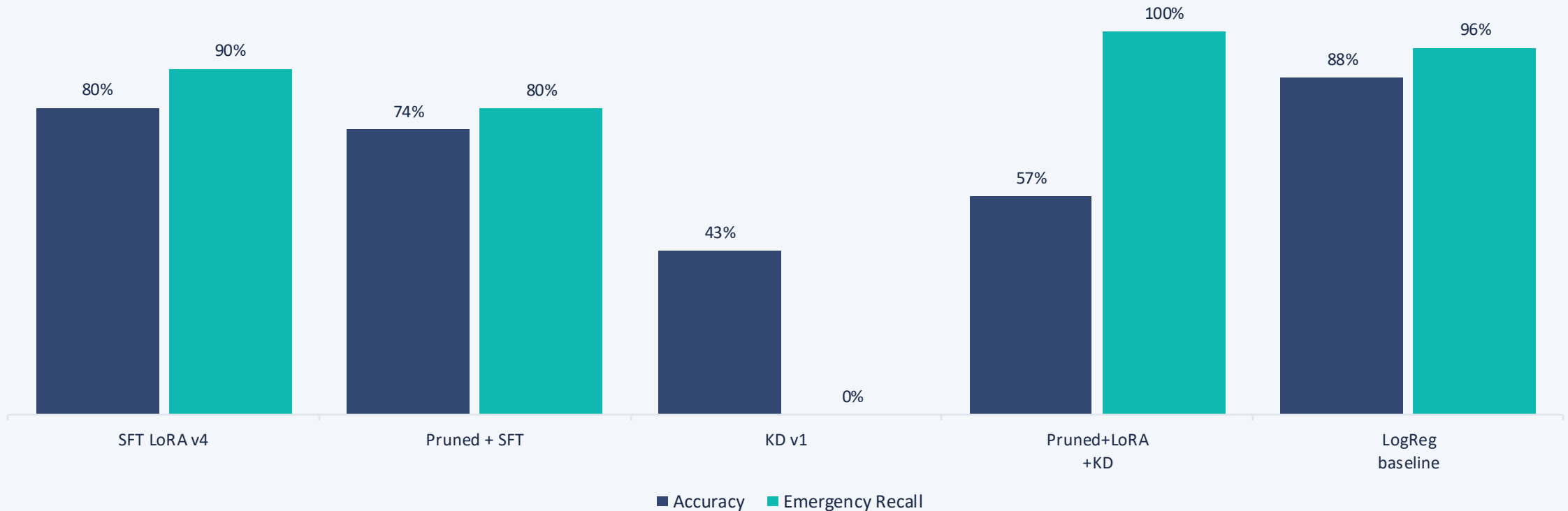
Results: Did We Hit the Targets?



Fine-tuned and pruned+LoRA both clear the bar. Pure distillation falls far short in its default decision mode — more on that shortly.

The Honest External Test: Real Hospital Data (MIMIC)

Accuracy vs. Emergency Recall on MIMIC-IV-ED



Surprise finding: a simple TF-IDF + Logistic Regression baseline matches or beats every neural model here — a humbling, useful result.

0.6B Beats 8B



13.3×

fewer parameters



10.6×

less VRAM at inference



4 / 4

test sets won outright



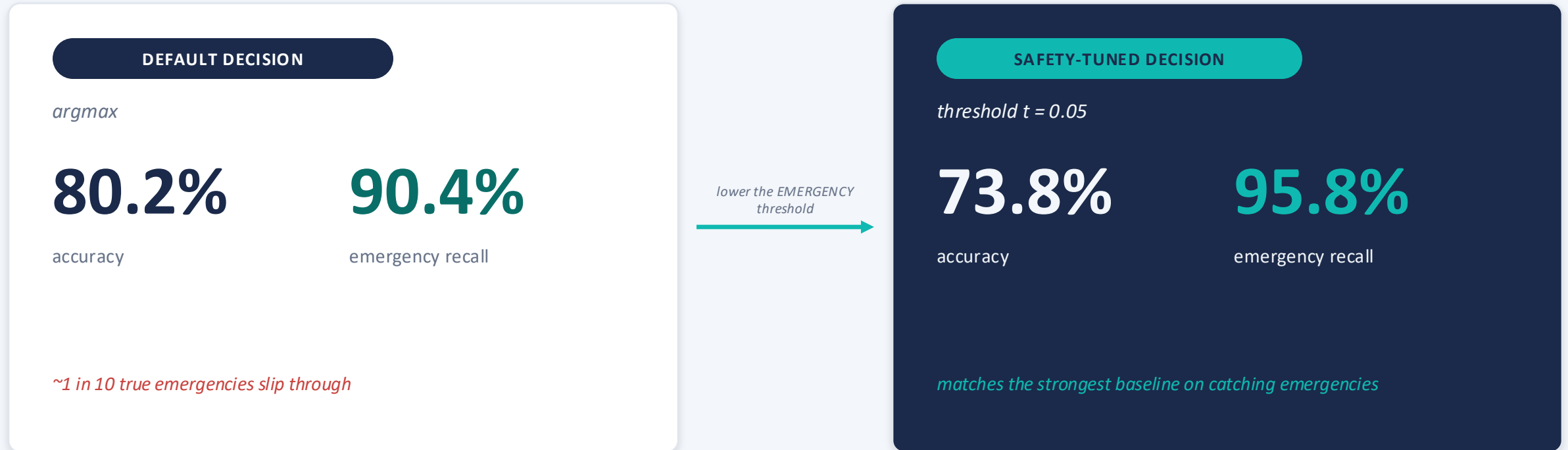
96.1%

teacher's parse-failure rate on real
notes

Student v4 outperforms the 8B teacher on every test set. The teacher reasons well but doesn't reliably follow the output format on messy real-world clinical text — the tuned student does both.

The Real Trade-off: Accuracy vs. Missed Emergencies

Same fine-tuned model — only the decision threshold for “EMERGENCY” changes:



In triage, a missed EMERGENCY is far costlier than a false alarm — so the decision threshold is tuned for recall, not raw accuracy.

Code Structure & Engineering Practices

src/	
└─ data_generation/	synthetic data + teacher inference
└─ pruning/	importance scoring, head + layer pruning
└─ distillation/	KD trainers (v1, v2), loss helpers
└─ finetuning/	LoRA fine-tune script
└─ approach2/	data merge, filtering, evaluation
└─ evaluation/	per-dataset eval + threshold sweeps
└─ demo/	Gradio chatbot demos

+ notebooks/, requirements.txt, README.md, app_streamlit.py

- **One script per pipeline stage** — modular, independently runnable via CLI arguments.
- **Versioning kept in the code itself:** kdtrainer.py → kdtrainer_v2.py, with comments documenting exactly what changed and why.
- **Large artifacts gitignored** — model weights and generated datasets never bloat the repo.
- **Two live demo apps** (Gradio + Streamlit) for interactive, qualitative testing beyond the metrics.

Built with

PyTorch

Transformers

PEFT / LoRA

bitsandbytes

datasets

scikit-learn

Gradio

Streamlit

Limitations & Honest Challenges



Pruning is “soft”

Weights are zeroed, not structurally removed — the architecture config is unchanged, so files barely shrink and inference isn't actually faster.



Distillation took three tries

KD v1 avoided EMERGENCY almost entirely; a full fine-tune attempt then collapsed outright. The debugging story is in the appendix.



Pruned generation breaks

Classification signal survives compression, but fluent free-form text generation degrades badly — confirmed on both 4-bit and fp16.



“Domain shift” isn't quite real

The Latvia test set shares standardized complaint vocabulary with training data; MIMIC is the tougher, more honest external test.



Format sensitivity

Both teacher and student fail to parse 85% of free-text, unstructured clinical case narratives from an out-of-distribution probe.



A simple baseline wins

TF-IDF + Logistic Regression matched or beat every neural model on emergency recall on real hospital data.

Where This Goes Next



Re-tune distillation (KD v3)

Lower the EMERGENCY class weight from 3.0 toward ~ 1.5 to fix v2's over-correction.



True structural pruning

Actually shrink weight matrices and the model config, not just zero out entries.



Finish deployment

4-bit / GGUF export and a local Ollama demo — planned but not yet completed.



Broader LoRA sweep

Rank, alpha, dropout, target modules, and learning rate, systematically compared.



A genuinely out-of-domain test

Real-world data that doesn't share standardized triage vocabulary with training.



RAG & preference tuning

Retrieval-augmented inference and DPO are scoped in the project plan but not yet run.

Key Takeaways

1

Good data + simple fine-tuning beat fancy compression.

LoRA on curated data outperformed both pruning and distillation here.

2

Size isn't everything.

A 13× smaller, specialized student beat its 8B teacher on every real test.

3

Optimize for the metric that matters.

For triage, that's Emergency Recall — not raw accuracy.

4

Test on truly external data.

The “easy” benchmark and the “honest” one told very different stories.

5

Negative results still count.

Two documented failure modes taught the team as much as the wins did.

Validating the Synthetic Reasoning Data



To validate the quality of the teacher-generated synthetic clinical reasoning chains, their semantic coherence was measured against real medical literature from PubMedQA, using RoBERTa-large BERTScore.

Class	Precision	Recall	F1-Score
EMERGENCY	81.1%	82.8%	81.9%
URGENT	81.1%	82.8%	81.9%
ROUTINE	81.4%	82.7%	82.0%
Overall	81.2%	82.7%	82.0%

BERTScore F1 around 0.82 across every class indicates the synthetic reasoning stays semantically close to real clinical literature — support for treating it as trustworthy training signal, not just plausible-sounding text.

Before Any Tuning: Raw Student vs. Raw Teacher

Dataset	Model	Total	Failed / Unparsed	Acc. (Valid)	Eff. Accuracy	Macro F1	EM Recall
Synthetic Test	Student	510	485 (95.1%)	96.0%	4.7%	0.327	0.0%
Synthetic Test	Teacher	510	350 (68.6%)	49.4%	15.5%	0.343	88.0%
Syntech-500	Student	500	493 (98.6%)	0.0%	0.0%	0.000	0.0%
Syntech-500	Teacher	500	53 (10.6%)	47.7%	42.6%	0.308	91.8%
Latvia (Real)	Student	3,000	2,976 (99.2%)	62.5%	0.5%	0.528	100.0%
Latvia (Real)	Teacher	3,000	1,585 (52.8%)	25.6%	12.1%	0.248	100.0%
MIMIC (Real)	Student	207	176 (85.0%)	58.1%	8.7%	0.286	90.0%
MIMIC (Real)	Teacher	207	199 (96.1%)	62.5%	2.4%	0.278	83.3%

The pattern: most raw-baseline “errors” are parsing failures, not reasoning failures — both models often can't be coaxed into the required output format at all. That's the gap fine-tuning closes.

Full Picture: Every Method, Every Dataset

Synthetic Test

State	Accuracy	EM Recall
Raw / Zero-Shot	4.7%	0.0%
Fine-Tuned (SFT)	100.0%	100.0%
Pruned SFT	100.0%	100.0%
Distilled (KD · t=0.05 sweep)	29.4%	100.0%

Syntech-500

State	Accuracy	EM Recall
Raw / Zero-Shot	0.0%	0.0%
Fine-Tuned (SFT)	96.2%	94.3%
Pruned SFT	87.0%	94.8%
Distilled (KD · t=0.05 sweep)	46.0%	100.0%

Latvia (Real)

State	Accuracy	EM Recall
Raw / Zero-Shot	0.5%	100.0%
Fine-Tuned (SFT)	100.0%	100.0%
Pruned SFT	100.0%	100.0%
Distilled (KD · t=0.05 sweep)	33.3%	100.0%

MIMIC (Real)

State	Accuracy	EM Recall
Raw / Zero-Shot	8.7%	100.0%
Fine-Tuned (SFT)	80.2%	90.4%
Pruned SFT	74.4%	80.0%
Distilled (KD · t=0.05 sweep)	57.1%	100.0%

Synthetic and Latvia are too easy to tell methods apart — Fine-Tuned and Pruned both hit 100%. Syntech-500 and MIMIC show where the real separation is.

The KD Journey: Three Attempts to Get It Right

ATTEMPT 1



KD v1

Plain student · logit KD
 $\alpha = 0.7, T = 4.0$

Teacher's EMERGENCY-averse soft labels dominate — student learns to avoid predicting EMERGENCY almost entirely.

~0% argmax recall

ATTEMPT 2



Full Fine-Tune

All 0.6B params, no LoRA
2 epochs, 42k samples

Catastrophic mode collapse — model predicts the majority ROUTINE class for nearly everything and generates gibberish text.

Effectively unusable

ATTEMPT 3



Pruned + LoRA + KD

LoRA adapter only, frozen pruned base
 $\alpha = 0.5, T = 2.0, EM \text{ weight} = 3.0$

No more collapse — but now input-invariant: predicts EMERGENCY almost regardless of symptoms (<5% spread across very different cases).

57.1% acc / 100% recall @ argmax

Pruned + LoRA + KD: Full Results



Once training was stable (previous slide), the argmax decision leans heavily toward EMERGENCY — high recall, lower accuracy. Sweeping the decision threshold recovers accuracy while holding recall near-perfect.

Dataset	Argmax Acc.	Argmax Recall	Best Threshold	Sweep Acc.	Sweep Recall
Synthetic Test	29.6%	100.0%	t = 0.65	70.2%	100.0%
Syntech-500	46.0%	100.0%	t = 0.55	62.4%	100.0%
Latvia (Domain-Shift)	33.3%	100.0%	t = 0.70	71.6%	100.0%
MIMIC (Real-World)	57.1%	100.0%	t = 0.80	61.9%	83.3%
PMC-Patients	0% parsed	0% parsed	N/A	0% parsed	0% parsed

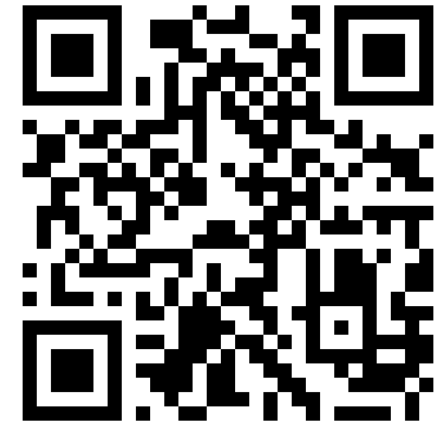
The PMC-Patients row isn't a data gap — it's the next challenge, on the following slide.

Live Demo — Try It Yourself



The fine-tuned student runs as an interactive triage chatbot — type in a patient complaint and watch it reason to an EMERGENCY / URGENT / ROUTINE call in real time.

<https://e9ad021fdd1d733c68.gradio.live/>



Scan to try it on your phone

(link above works too)



Questions?

13.3×

smaller than teacher

90.8%

triage accuracy

95.8%

emergency recall (tuned)